

Question number	Answer	Additional guidance	Mark
8(a)	An explanation that includes four of the following points: • rhodopsin (in rod cells) absorb light and {splits into retinal and opsin / becomes bleached}(1) • cis-retinal converted to trans-retinal(1) • sodium ion channels (in surface membrane) close (1) • {rod cell / membrane} becomes hyperpolarised (1) • no glutamate transmitter {released/produced} (into synaptic cleft) (1)	accept rhodopsin absorbs light and forms opsin /light hits rhodopsin and bleaches it accept membrane permeability to sodium ions decreases do not accept (opsin) blocks Na⁺ channel	Expert (4)

Question number	Answer	Additional guidance	Mark
8(b)	An answer that includes three of the following points: • extraction of mRNA (1) • use of (specific) microarray (1) • use of fluorescent {dyes /marker/ probe}(1) • (comparing stimulated and un-stimulated neurones) {use of bioinformatics/ algorithms to analyse data} (1) OR • extraction of mRNA(1) • use of reverse transcriptase to form cDNA(1) • use of PCR(1) • use of gel electrophoresis and compare banding(1)	Do not pick and mix accept reference to samples of mRNA [ignore marker gene]	Expert (3)
Question number	Answer	Additional guidance	Mark
8(c)	An explanation that includes three of the following points: • (capsaicin){activates/binds to} a receptor (1) • opens ion channels in neurons (1) • results in {depolarization / action potential / impulse} (in neuron (1) • relay neuron transmits {action potential/ impulse} (to the brain where) it is perceived as pain (1)	accept TRPV1 as named receptor accept named ion channel do not accept if in motor neurone	Expert (3)

Question number	Answer	Additional guidance	Mark
8(d)	An explanation that includes three of the following • (stimulus causes) ion channel to change shape (1) • (which causes) ion channels {to / are} open(1) • allowing influx of (sodium / calcium) ions (1) • initiates {depolarization/ action potential} (1)	ignore potassium ions accept calcium ions cause release of neurotransmitters accept if it goes above threshold level	Expert (3)

Question number	Answer	Additional guidance	Mark
8(e)	A explanation that includes three of the following points: • {mechanical stimulus / pressure} changes shape of the membrane (1) • causes {ion channel / protein} to {change shape / activated / open} (1) • allowing influx of (sodium / calcium) ions (1) • initiates {depolarization/ action potential} (1)	accept stretches membrane ignore potassium ions	Expert (3)

Question number	Answer	Additional guidance	Mark
8(f)	An explanation that includes two of the following points: • by comparing {DNA base/amino acid} sequences (1) • piezo 2 has similar structure to piezo 1(1) • both Piezo1 and Piezo 2(gene) could be inactivated in the same way (1) • both (Piezo1 and Piezo2) are activated by pressure (on cell membrane) (1)	accept a second protein / amino acid with a similar {structure / DNA base sequence} was identified accept pressure makes them both open accept respond to mechanical stimuli accept respond the same to stimuli	Expert (2)

Question number	Answer	Additional guidance	Mark
8(g)	An answer that includes three of the following points: • TRPV1 channels are sensitive to temperature (of the blood) (1) • impulse transmitted to {CNS / ANS / hypothalamus/brain} (1) • credit description of what (hypothalamus does) to maintain core body temperature (1)		Expert (2)

